

- Q.1₃₃ Whenever horses a, b, c race together, their respective probabilities of winning the race are 0.3, 0.5 and 0.2 respectively. If they race three times the probability that “the same horse wins all the three races” and the probability that a, b, c each wins one race, are respectively

(A*) $\frac{8}{50}; \frac{9}{50}$ (B) $\frac{16}{100}, \frac{3}{100}$ (C) $\frac{12}{50}; \frac{15}{50}$ (D) $\frac{10}{50}; \frac{8}{50}$

- [Sol. $P(a) = 0.3$; $P(b) = 0.5$; $P(c) = 0.2 \Rightarrow a, b, c$ are exhaustive
 $P(\text{same horse wins all the three races}) = P(aaa \text{ or } bbb \text{ or } ccc)$

$$= (0.3)^3 + (0.5)^3 + (0.2)^3 = \frac{27+125+8}{1000} = \frac{160}{1000} = \frac{4}{25}$$

$P(\text{each horse wins exactly one race})$

$$= P(abc \text{ or } acb \text{ or } bca \text{ or } bac \text{ or } cab \text{ or } cba) = 0.3 \times 0.5 \times 0.2 \times 6 = 0.18 = \frac{9}{50}]$$

- Q.2₆₃ Let A & B be two events. Suppose $P(A) = 0.4$, $P(B) = p$ & $P(A \cup B) = 0.7$. The value of p for which A & B are independent is :

(A) $1/3$ (B) $1/4$ (C*) $1/2$ (D) $1/5$

- [Sol. $P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$

$$0.7 = 0.4 + p - 0.4p$$

$$\therefore 0.6p = 0.3 \Rightarrow p = \frac{1}{2}]$$

- Q.3₇₈ A & B are two independent events such that $P(\bar{A}) = 0.7$, $P(\bar{B}) = a$ & $P(A \cup B) = 0.8$, then, a =

(A) $5/7$ (B*) $2/7$ (C) 1 (D) none

- Q.4₇₂ A pair of numbers is picked up randomly (without replacement) from the set $\{1, 2, 3, 5, 7, 11, 12, 13, 17, 19\}$. The probability that the number 11 was picked given that the sum of the numbers was even, is nearly :

(A) 0.1 (B) 0.125 (C*) 0.24 (D) 0.18

- [Hint: $P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{{}^7C_1}{{}^8C_2+1} = \frac{7}{29}$; A : 11 is picked, B : sum is even]

- Q.5₄₂ For a biased die the probabilities for the different faces to turn up are given below :

Faces :	1	2	3	4	5	6
Probabilities :	0.10	0.32	0.21	0.15	0.05	0.17

The die is tossed & you are told that either face one or face two has turned up. Then the probability that it is face one is :

(A) $1/6$ (B) $1/10$ (C) $5/49$ (D*) $5/21$

- [Hint: $P(A/A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A) + P(B)} = \frac{0.10}{0.10 + 0.32}]$

- Q.6₁₇₀ A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability that the determinant chosen has the value non negative is :
 (A) 3/16 (B) 6/16 (C) 10/16 (D*) 13/16

[Hint: $1 - P(\text{Determinant has negative value})$]

$$1 - \frac{3}{16} = \frac{13}{16} \left(\left| \begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array} \right| ; \left| \begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array} \right| ; \left| \begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right| \right)]$$

- Q.7₉₉ 15 coupons are numbered 1, 2, 3, ..., 15 respectively. 7 coupons are selected at random one at a time with replacement. The probability that the largest number appearing on a selected coupon is 9 is :

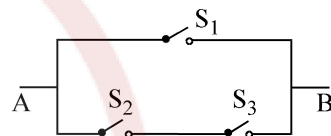
(A) $\left(\frac{9}{16}\right)^6$ (B) $\left(\frac{8}{15}\right)^7$ (C) $\left(\frac{3}{5}\right)^7$ (D*) $\frac{9^7 - 8^7}{15^7}$

[Hint: $n(S) = \times \times \times \times \times \times \times = 15^7$; came of the number must be 9]

- Q.8₂₂₉ A card is drawn & replaced in an ordinary pack of 52 playing cards. Minimum number of times must a card be drawn so that there is atleast an even chance of drawing a heart, is
 (A) 2 (B*) 3 (C) 4 (D) more than four

[Hint: $\frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{1}{4}$]

- Q.9 An electrical system has open-closed switches S_1 , S_2 and S_3 as shown.



The switches operate independently of one another and the current will flow from A to B either if S_1 is closed or if both S_2 and S_3 are closed. If $P(S_1) = P(S_2) = P(S_3) = \frac{1}{2}$, find the probability that the circuit will work.

[Sol. $P(S_1) = P(S_2) = P(S_3) = \frac{1}{2}$ [Ans. $\frac{5}{8}$]

E : event that the current will flow.

$$P(E) = P((S_2 \cap S_3) \text{ or } S_1) = P(S_2 \cap S_3) + P(S_1) - P(S_1 \cap S_2 \cap S_3) = \frac{1}{4} + \frac{1}{2} - \frac{1}{8} = \frac{5}{8}$$

- Q.10 A certain team wins with probability 0.7, loses with probability 0.2 and ties with probability 0.1. The team plays three games. Find the probability

(i) that the team wins at least two of the games, but lose none.

(ii) that the team wins at least one game.

[Ans. (i) 0.49 ; (ii) 0.973]

[Sol. $P(W) = 0.7$; $P(L) = 0.2$; $P(T) = 0.1$

E : winning at least 2 games but lose none

$$P(E) = P(W W T \text{ or } W T W \text{ or } T W W \text{ or } W W W)$$

$$= 3 \times 0.7 \times 0.7 \times 0.1 + (0.7)^3 = 0.7 \times 0.7 [0.3 + 0.7] = 0.49$$

F : wining at least 1 game

$$A = L \text{ or } T \Rightarrow P(A) = 0.3 ; P(F) = 1 - P(A A A) = 1 - (0.3)^3 = 1 - 0.027 = 0.973]$$

Q.11 An integer is chosen at random from the first 200 positive integers. Find the probability that the integer is divisible by 6 or 8.

[Sol. $n(S) = 200$

[Ans. $\frac{1}{4}$]

$n(A) = \text{divisible by 6 or 8}$

$$n(6) = 6 + 12 + \dots + 198 = 33$$

$$n(8) = 8 + 16 + \dots + 200 = 25$$

$$n(6 \cap 8) = 24 + 48 + \dots + 192 = 8$$

$$n(A) = 33 + 25 - 8 = 50$$

$$P(A) = \frac{1}{4} \quad]$$

Q.12 A clerk was asked to mail four report cards to four students. He addresses four envelopes that unfortunately paid no attention to which report card be put in which envelope. What is the probability that exactly one of the students received his (or her) own card?

[Ans. $\frac{8}{24} = \frac{1}{3}$]

Q.13 Find the probability of at most two tails or at least two heads in a toss of three coins.

[Sol. $A = \text{at most two tails}$ $n(s) - \{T T T\}$ [Ans. $\frac{7}{8}$]

$B = \text{at least two heads}$ $H H H, T H H, H T H, H H T$

$$P(A) = \frac{7}{8}; P(B) = \frac{4}{8}; P(A \cap B) = \frac{4}{8}$$

$$P(A \cup B) = \frac{7}{8} + \frac{4}{8} - \frac{4}{8} = \frac{7}{8} \quad \text{Ans.]}$$

Q.14 What is the probability that in a group of

(i) 2 people, both will have the same date of birth.

(ii) 3 people, at least 2 will have the same date of birth.

Assume the year to be ordinary consisting of 365 days.

[Sol.(i) A : both have the same date of birth

$$P(A) = \frac{1}{365} \cdot \frac{1}{365} + \frac{1}{365} \cdot \frac{1}{365} + \dots 365 \text{ times} = \frac{1}{365} \quad \text{Ans.}$$

(ii) B : at least 2 in a group of 3 will have the same birthday

$$P(B) = 1 - P(\text{all 3 have different birthday})$$

$$= 1 - 1 \times \frac{364}{365} \cdot \frac{363}{365} = 1 - \frac{364 \times 363}{(365)^2} \quad \text{Ans.]}$$

Q.15 The probability that a person will get an electric contract is $\frac{2}{5}$ and the probability that he will not get plumbing contract is $\frac{4}{7}$. If the probability of getting at least one contract is $\frac{2}{3}$, what is the probability that he will get both?

[Sol. $P(E) = \frac{2}{5}; P(F) = P(\text{plumbing}) = 1 - \frac{4}{7} = \frac{3}{7}$

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

$$\frac{2}{3} = \frac{2}{5} + \frac{3}{7} - x \quad \Rightarrow \quad x = \frac{17}{105} \quad \text{Ans.]}$$

Q.16 Five horses compete in a race. John picks two horses at random and bets on them. Find the probability that John picked the winner. Assume dead heat.

[Sol. $n(S) = {}^5C_2 = 10$
 $n(A) = 1 \cdot {}^4C_1 = 4$] $\Rightarrow p = \frac{2}{5}$] [Ans. $\frac{4}{10}$]

Q.17 Two cubes have their faces painted either red or blue. The first cube has five red faces and one blue face. When the two cubes are rolled simultaneously, the probability that the two top faces show the same colour is $1/2$. Number of red faces on the second cube, is

- (A) 1 (B) 2 (C*) 3 (D) 4

[Sol. Let the number of red faces on the 2nd cube = x
 number of blue faces = (6 - x)

$$P(RR \text{ or } BB) = 1/2$$

$$\frac{5}{6} \cdot \frac{x}{6} + \frac{1}{6} \cdot \frac{6-x}{6} = \frac{1}{2}$$

$$5x + 6 - x = 18$$

$$4x = 12 \Rightarrow x = 3 \text{ Ans.]}$$

Q.18 A H and W appear for an interview for two vacancies for the same post.

$P(H) = 1/7$; $P(W) = 1/5$. Find the probability of the events

- (a) Both are selected (b) only one of them is selected (c) none is selected.

[Ans. $\frac{1}{35}, \frac{2}{7}, \frac{24}{35}$]

Q.19 A bag contains 6R, 4W and 8B balls. If 3 balls are drawn at random determine the probability of the event

- (a) all 3 are red ; (b) all 3 are black ; (c) 2 are white and 1 is red ;
 (d) at least 1 is red ; (e) 1 of each colour are drawn
 (f) the balls are drawn in the order of red, white, blue.

[Ans. (a) $\frac{5}{204}$, (b) $\frac{7}{102}$, (c) $\frac{3}{68}$, (d) $\frac{149}{204}$, (e) $\frac{4}{17}$, $\frac{2}{51}$]

Q.20 The odds that a book will be favourably reviewed by three independent critics are 5 to 2, 4 to 3, and 3 to 4 respectively. What is the probability that of the three reviews a majority will be favourable?

[Ans. $\frac{209}{343}$]

Q.21 In a purse are 10 coins, all five nP's except one which is a rupee, in another are ten coins all five nP's. Nine coins are taken from the former purse and put into the latter, and then nine coins are taken from the latter and put into the former. Find the chance that the rupee is still in the first purse.

[Ans. $\frac{10}{19}$]

Q.22 A, B, C in order cut a pack of cards, replacing them after each cut, on condition that the first who cuts a spade shall win a prize. Find their respective chances.

[Ans. $\frac{16}{37}, \frac{12}{37}, \frac{9}{37}$]

Q.23 A and B in order draw from a purse containing 3 rupees and 4 nP's, find their respective chances of first drawing a rupee, the coins once drawn not being replaced.

[Ans. $\frac{22}{35}, \frac{13}{35}$]